



Incident report analysis

Summary	The organization experienced a Denial of Service (DoS) attack that disrupted internal network operations for approximately two hours. The attack was caused by a flood of incoming ICMP packets that overwhelmed network resources and prevented legitimate users from accessing network services. Investigation revealed that the attacker exploited an unconfigured firewall, which allowed excessive ICMP traffic to enter the network. The incident response team mitigated the attack by blocking incoming ICMP traffic, temporarily shutting down non-critical services, and restoring critical services. The security team later implemented firewall rate limiting, source IP verification, network monitoring software, and an IDS/IPS solution to reduce the risk of similar attacks in the future.
Identify	The attack was identified as a Denial of Service (DoS) attack using ICMP flood traffic. The primary systems affected were the organization's internal network infrastructure, network services, and resources that depended on network connectivity. The root cause was an improperly configured firewall that failed to restrict incoming ICMP traffic. This vulnerability allowed the attacker to overwhelm network resources and disrupt business operations.
Protect	To strengthen security, the organization should maintain firewall rules that limit ICMP traffic and regularly review firewall configurations. Source IP verification should remain enabled to identify spoofed packets. The company should establish a formal network hardening policy, perform regular security audits, apply configuration management procedures, and provide security awareness training for IT personnel responsible for firewall administration. Additional protections should include network segmentation and periodic vulnerability

	assessments.
Detect	The organization should use continuous network monitoring software to identify abnormal traffic patterns and unusual spikes in ICMP traffic. IDS and IPS technologies should be configured to alert security personnel when suspicious traffic exceeds established thresholds. Firewall logs, network traffic analysis tools, and a SIEM solution should be used to collect and correlate security events. Regular monitoring of network performance metrics will help identify potential attacks before they significantly affect operations.
Respond	If a similar attack occurs, the incident response team should immediately activate the incident response plan, identify affected systems, and contain the attack by filtering or blocking malicious traffic. Non-essential services may be temporarily isolated to preserve critical business operations. Security personnel should collect firewall logs, IDS/IPS alerts, and network traffic data for forensic analysis. Communication procedures should ensure that management, IT staff, and affected stakeholders receive timely updates. After the incident, the team should conduct a lessons-learned review and update response procedures as necessary.
Recover	The organization should restore normal network operations by verifying that malicious traffic has been eliminated and confirming that all critical services are functioning correctly. System configurations should be reviewed and validated before returning services to production. Recovery efforts should include restoring any affected services, validating network performance, reviewing security controls, and documenting recovery actions. Backup configurations and disaster recovery procedures should be tested regularly to ensure rapid restoration during future incidents.

Reflections/Notes:

This incident demonstrated the importance of proper firewall configuration, continuous monitoring, and layered security controls. Applying the NIST Cybersecurity Framework helped identify security gaps and develop a structured approach for protecting, detecting, responding to, and recovering from cybersecurity incidents. Future security efforts should focus on proactive monitoring, regular audits, and continuous improvement of network defenses.